Ideation Phase
Brainstorm & Idea Prioritization Template

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Step 1: Brainstorm and Idea Listing

Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Vishwas Choyal	Nine-view interactive Tableau dashboard	BI dashboard	1
2	Vishwas Choyal	Spreadsheet with static summaries	Spreadsheet	2
3	Vishwas Choyal	Battery comparison report	Research report	3
4	Vishwas Choyal	Machine-learning classifier	Predictive modelling	4

Step 2: Idea Prioritization

Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility (High/ Medium/Lo w)	Importance (High/ Medium/Lo w)	Priority	Selected (Yes/No)
1	Interactive Tableau dashboard	High	High	1	Yes
2	Static spreadsheet report	High	Medium	2	No
3	Predictive classifier	Medium	Low	3	No

Selection rationale

The dashboard directly addresses the three supplied scenarios and the nine-visualization deliverable. Classification would answer a different question and require a separate holdout evaluation.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Ideation Phase
Define the Problem Statements

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - *who are they?*	Describe the customer and their attributes here
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – *what bothers them most?*	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Describe the emotions the result from experiencing the problems or barriers

Reference: https://miro.com/templates/customer-problem-statement/

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Emily Johnson, market strategy director	Prioritize EV product segments	Sales, ratings and range are scattered	There is no shared view of segment performance	Uncertain about portfolio priorities
PS-2	David Chen, BI manager	Compare manufacturers and production geography	Totals and shares are hard to compare	Raw CSVs lack an interactive analytical view	Concerned about inconsistent client advice
PS-3	Dr Sarah Williams, researcher	Compare battery technology and mobility outcomes	Aggregate trends can conceal category differences	Technical metrics and adoption data are separated	Cautious about overinterpreting evidence

Evidence boundary

These personas come from the supplied project scenarios. They are not interview participants. This project analyzes a synthetic dataset and does not claim real-world market outcomes.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Ideation Phase
Empathize & Discover

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	4 Marks

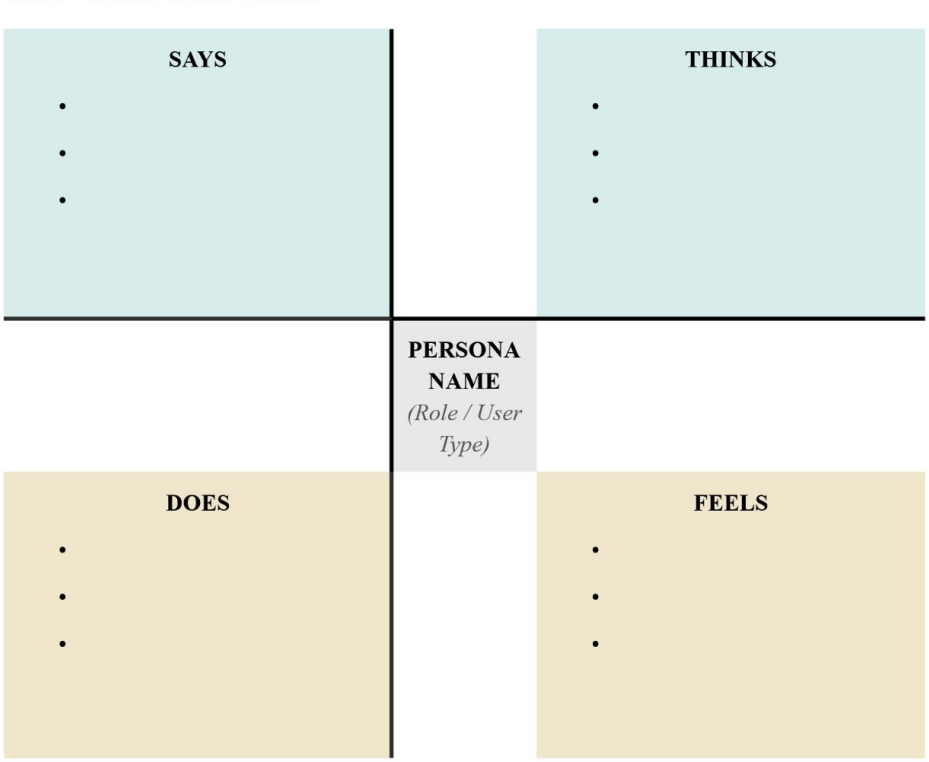
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:

Reference: https://www.mural.co/templates/empathy-map-canvas



Completed empathy canvas

Persona assumptions below are derived from the project overview, rather than customer interviews. The linked Mural template is retained as a reference attachment.

Canvas area	Scenario-based content
Who	Emily Johnson, EV market strategy director
Goal and tasks	Compare sales, satisfaction, battery adoption and range to prioritize product segments.
Hear	Scenario assumption: leadership requests evidence for product portfolio decisions.
See	A large CSV with manufacturer, category, battery and country fields.
Say	Scenario paraphrase: I need executive-level insight into market performance.
Do	Compare segments, select a category and inspect the linked charts.
Think and feel	Needs consistent metrics and clear boundaries on synthetic-data interpretation.

Canvas area	Scenario-based content
Pains	Missing measurements, fragmented comparisons and ambiguous model-year meaning.
Gains	One interactive dashboard with nine views and documented calculations.

Completed linked Mural template

The downloaded template is filled below. The preceding editable table provides the same responses at reading size.



Scenario-derived assumptions, not interviews. Source template: Mural empathy map canvas.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Customer Journey Map

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	2 Marks

Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of journey	Discover	Compare	Decide
Actions *What does the customer do?*	Read project overview; open dashboard.	Select category; compare nine views.	Review evidence; note limitations.
Touchpoint *What part of the service do they interact with?*	SkillWallet overview; Tableau Public.	Category bars; chart tooltips.	Report, workbook and demo.
Customer Thought *What is the customer thinking?*	Which questions must I answer?	Are metrics comparable and complete?	What does the evidence support?
Customer Feeling *What is the customer feeling?*	Curious; uncertain about raw data.	More informed; checking assumptions.	Clearer about dataset findings.

Process Ownership *Who is in the lead on this?*	Project author	Dashboard viewer	Reviewer
Opportunities *How can we improve this stage?*	State synthetic scope early.	Show units and null handling.	Link evidence and reproducible files.

Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).

Symbol	Name	Description
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Analyst	Data acquisition	US-01	Download both Kaggle CSV splits	300,000 source rows inventoried	High	Sprint 1

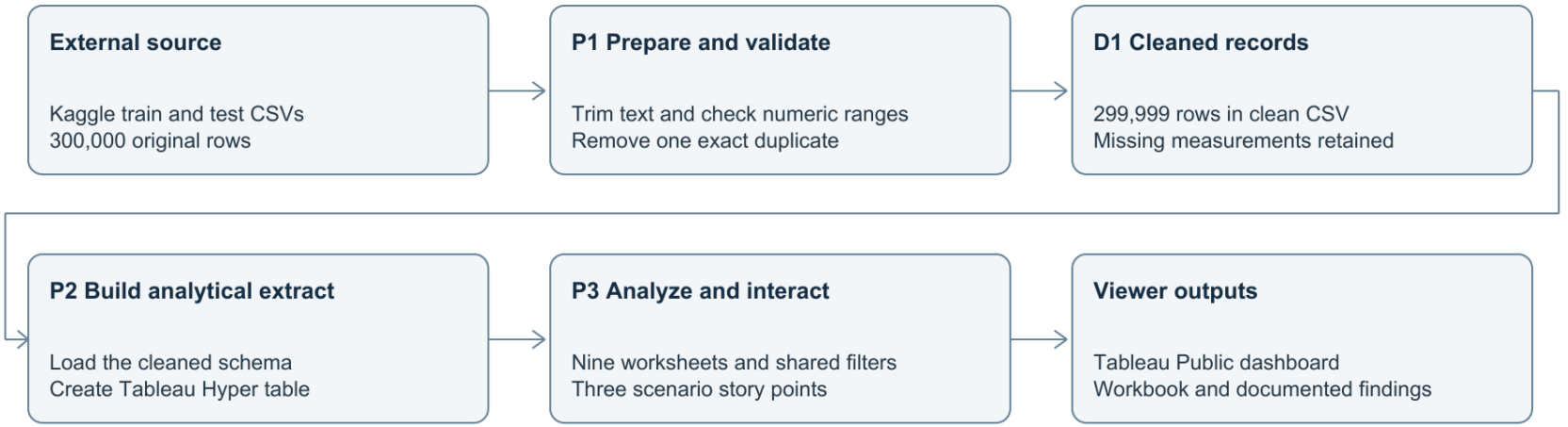
User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Analyst	Data preparation	US-02	Remove exact duplicates and preserve nulls	299,999 rows; one duplicate removed	High	Sprint 1
Emily	Market analysis	US-03	Compare category sales and ratings	Views 1 and 3 show sums and means	High	Sprint 1
Emily	Cohort analysis	US-04	Compare model-year segment composition	Views 2 and 9 label model year correctly	High	Sprint 1
David	Competition	US-05	Compare manufacturer totals and sales shares	View 4 has ten manufacturers and share tooltips	High	Sprint 1
David	Geography	US-06	Compare manufacturing-country sales	View 6 distinguishes manufacturing from market region	High	Sprint 1
Sarah	Technology	US-07	Compare battery mix, capacity and range	Views 5, 7 and 8 cover technology metrics	High	Sprint 1
Dashboard viewer	Interaction	US-08	Select a vehicle category to filter charts	Linked charts update and clear selection restores totals	High	Sprint 1
Mentor	Delivery	US-09	Review report, workbook and narrated demo	Public links and files can be opened	High	Sprint 2

Implemented data flow

Kaggle CSV files → Python validation and exact deduplication → cleaned CSV → Tableau Hyper extract → nine worksheets → interactive dashboard → Tableau Public and project documentation.

The raw files remain the source of truth. The published dashboard contains synthetic analytical data and no login credentials.

EV project data flow



Read left to right. Upper row feeds the lower row through the rightmost data store.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Load both EV dataset CSV splits and retain source provenance.	High
2	Data Processing	Trim text, remove one exact duplicate and preserve missing measurements.	High
3	Business Analysis	Calculate observed sales sums, mean ratings, mean range and shares of dataset sales.	High
4	Business Insights & Analytical Outputs	Provide nine views covering all three supplied scenarios.	High
5	User Interaction & Analysis	Selecting a category filters linked views; clearing restores all categories.	High
6	Data Update & Reporting	Rebuild the clean CSV and Hyper extract when source files change; republish the workbook.	High

Non-functional Requirements:
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Readable titles, units and consistent chart layout.
NFR-2	**Security**	Publish only synthetic data and project artifacts; exclude account secrets.
NFR-3	**Reliability**	Reconcile row count and total sales independently; retain null semantics.
NFR-4	**Performance**	Use a Hyper extract; verify category selection updates all target charts.
NFR-5	**Availability**	Portable TWBX for offline access; Tableau Public for browser access.

FR No.	Non-Functional Requirement	Description
NFR-6	**Scalability**	Aggregate for rendering and rebuild extracts as source volume increases.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python 3 and pandas	Repeatable validation and aggregation.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV and Tableau Hyper	Preserve source portability and accelerate queries.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Tableau text connector and Hyper API	Load the original schema and the cleaned extract.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	pandas and Tableau calculations	Deduplicate, preserve nulls and define analytical measures.

S.No	Component	Description	Technology	Justification / Purpose
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Public 2026.3	Build nine worksheets and an interactive dashboard.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public and GitHub	Browser access and a versioned project submission.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Project Design Phase-I
Problem – Solution Fit Template

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	2 Marks

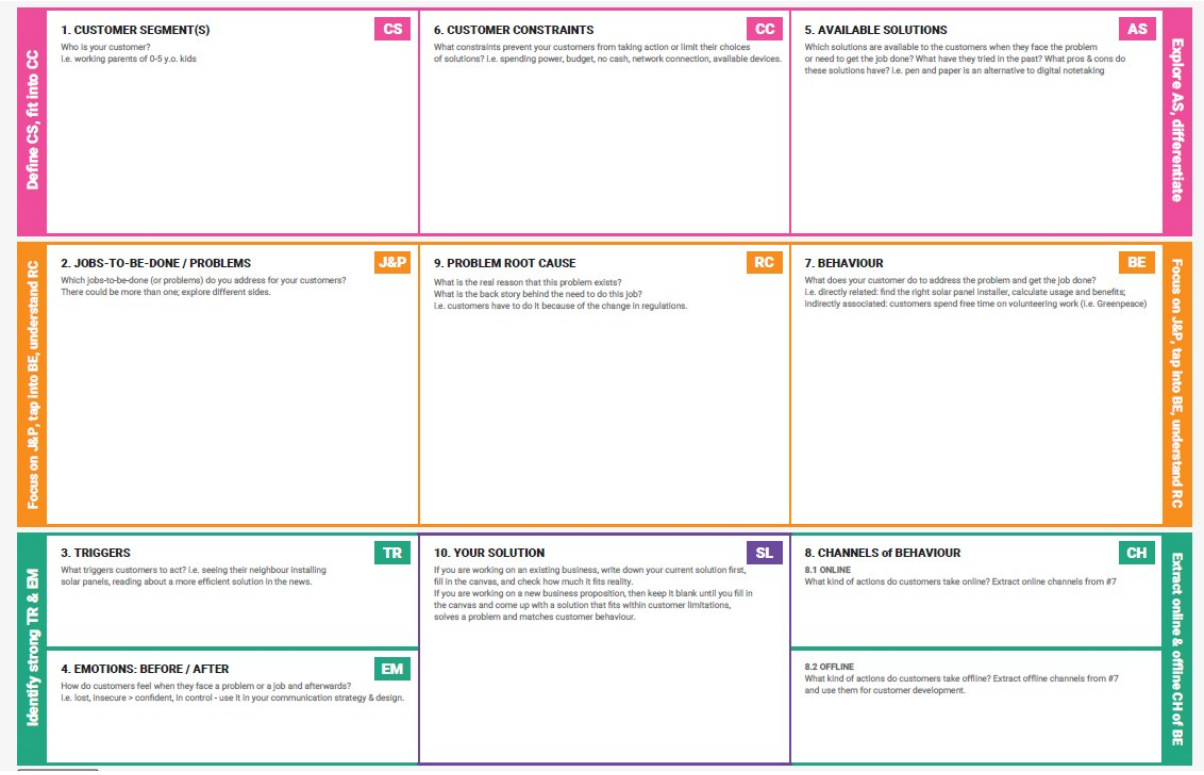
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/

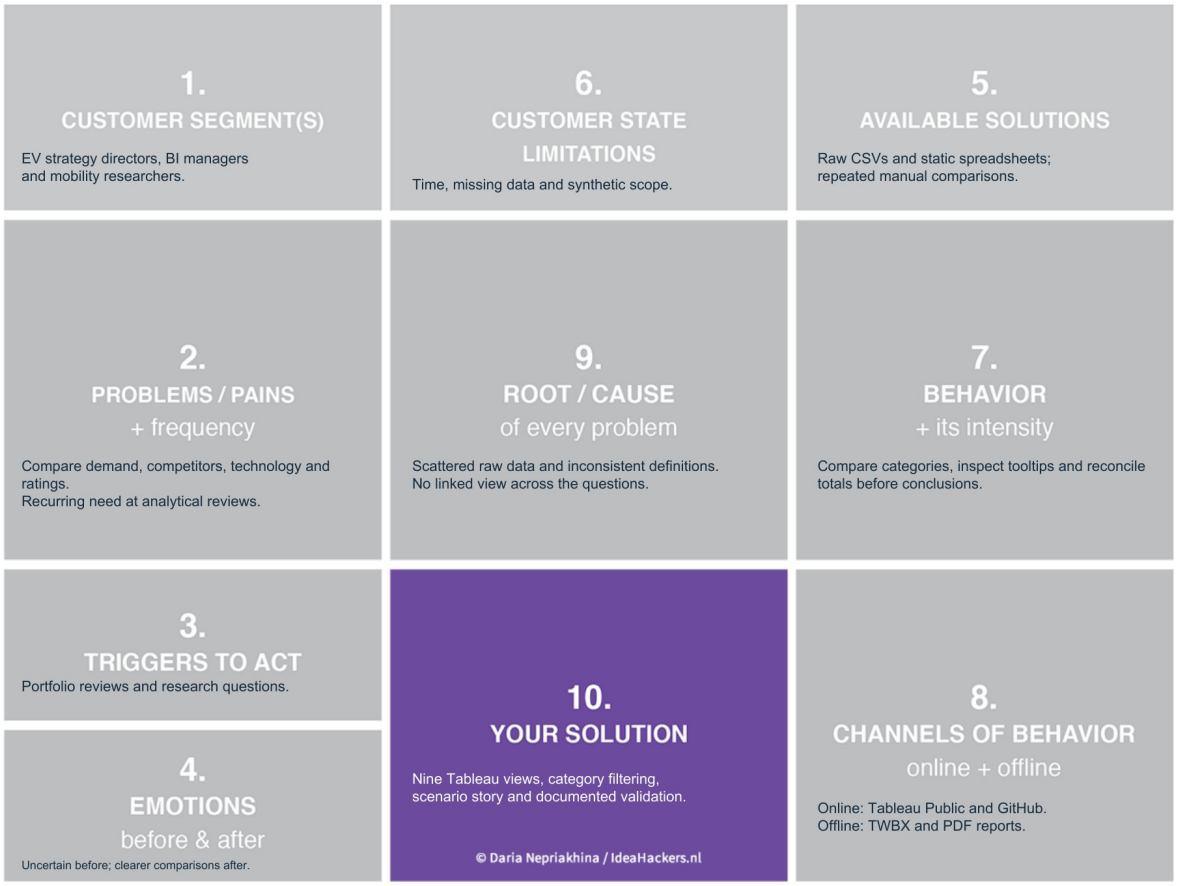
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

Canvas area	Completed project response
Customer segments	EV strategy directors, automotive BI consultants and sustainable mobility researchers.
Jobs and problems	Compare segment demand, manufacturer performance, battery metrics and satisfaction.
Triggers	Portfolio reviews, client market-entry questions and technology research.
Emotions before and after	Scenario assumption: uncertainty from fragmented data; greater clarity from defined, comparable measures.
Available solutions	Raw CSVs and static spreadsheets; flexible but require repeated manual comparisons.
Constraints	Time, analytical skill, missing measurements and synthetic-data scope.
Behaviour	Inspect rankings, compare segments and validate totals before drawing conclusions.
Online channels	Tableau Public dashboard and linked project repository.
Offline channels	Portable TWBX and PDF documentation for review.
Root cause	Unstructured analytical workflow and absent shared definitions.
Solution	Nine-view Tableau dashboard, category filter and documented data-quality checks.

Completed linked IdeaHackers template

The downloaded canvas is completed below; original template attribution is retained.

EV project problem solution fit canvas



Scenario-derived responses. Source: IdeaHackers canvas by Daria Nepriakhina; original attribution retained.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

SMARTBRIDGE
Let's Bridge the Gap

Project Initialization and Planning Phase

Date	31 October 2026
Team ID	Individual
Project Title	Electric Vehicle Market Performance and Insights
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Answer the three supplied EV analysis scenarios with an interactive Tableau dashboard containing nine visualizations.
Scope	Descriptive analysis of 299,999 cleaned synthetic records across model years 2018–2026. No predictive model or real-market forecasts.
Problem Statement	
Description	Users need comparable information on segments, manufacturers, batteries, geography and customer ratings.
Impact	Reduce manual CSV comparisons and make assumptions and metric definitions visible.
Proposed Solution	
Approach	Acquire both CSVs, validate and deduplicate, build a Hyper extract, develop nine views, verify filtering and document the evidence.
Key Features	Category filtering, manufacturer ranking, battery comparisons, cohort heatmap and portable workbook.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Windows PC capable of running Tableau Public; no GPU computation required.
Memory	RAM specifications	8 GB minimum, 16 GB recommended.
Storage	Disk space for data, models, and logs	At least 2 GB available for data and deliverables.
Software		
Frameworks	Python frameworks	Tableau Public desktop authoring.
Libraries	Additional libraries	pandas, Tableau Hyper API and document-generation libraries.
Development Environment	IDE, version control	Local scripts and Tableau Public; GitHub for delivery.
Data		
Data	Source, size, format	Kaggle synthetic EV CSVs, 300,000 raw rows and 37 original fields.

Project author
Vishwas Choyal | EN25CS3011066 | Individual project

Project Design Phase-I
Solution Architecture

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Implemented component architecture

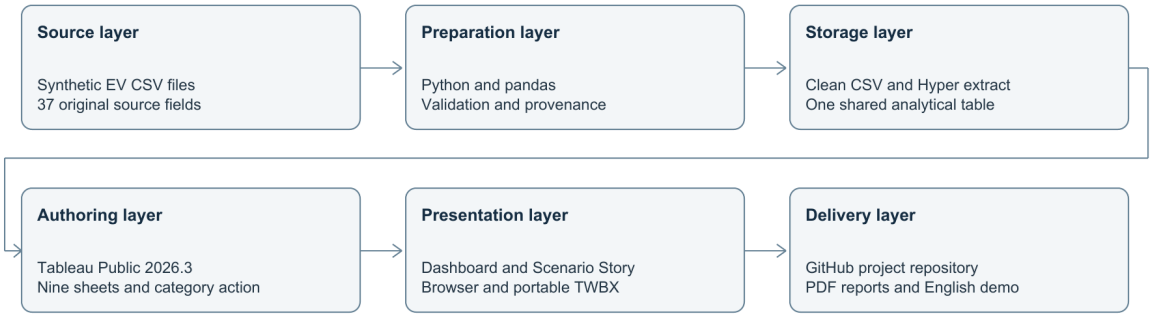
Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Source	200,000 train and 100,000 test rows	Kaggle CSV
Preparation	Null audit, type checks and exact deduplication	Python and pandas
Analytical store	299,999 validated records	Tableau Hyper
Visualization	Nine worksheets and category filter action	Tableau Public
Delivery	Workbook, documents and narrated demo	GitHub and Google Drive

Project architecture

Source files feed a repeatable preparation step. The cleaned data feeds one Hyper extract shared by all nine Tableau worksheets. The dashboard applies a category-selection filter to linked sheets and is delivered as a self-contained TWBX.

EV dashboard solution architecture



Read left to right. Upper row feeds the lower row through the rightmost data store.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint 1	Acquisition	US-01	Download source and inspect schema	2	High	Vishwas Choyal
Sprint 1	Preparation	US-02	Validate and deduplicate records	3	High	Vishwas Choyal
Sprint 1	Visualization	US-03	Build nine charts	5	High	Vishwas Choyal
Sprint 1	Dashboard	US-04	Verify category filtering	3	High	Vishwas Choyal
Sprint 1	Testing	US-05	Reconcile totals and verify portable file	2	High	Vishwas Choyal

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint 2	Documentation	US-06	Complete phase reports	3	High	Vishwas Choyal
Sprint 2	Demonstration	US-07	Record English narrated walkthrough	3	High	Vishwas Choyal
Sprint 2	Submission	US-08	Publish links and submit for mentor review	2	High	Vishwas Choyal

Project Tracker, Velocity & Burndown Chart: (4 Marks)

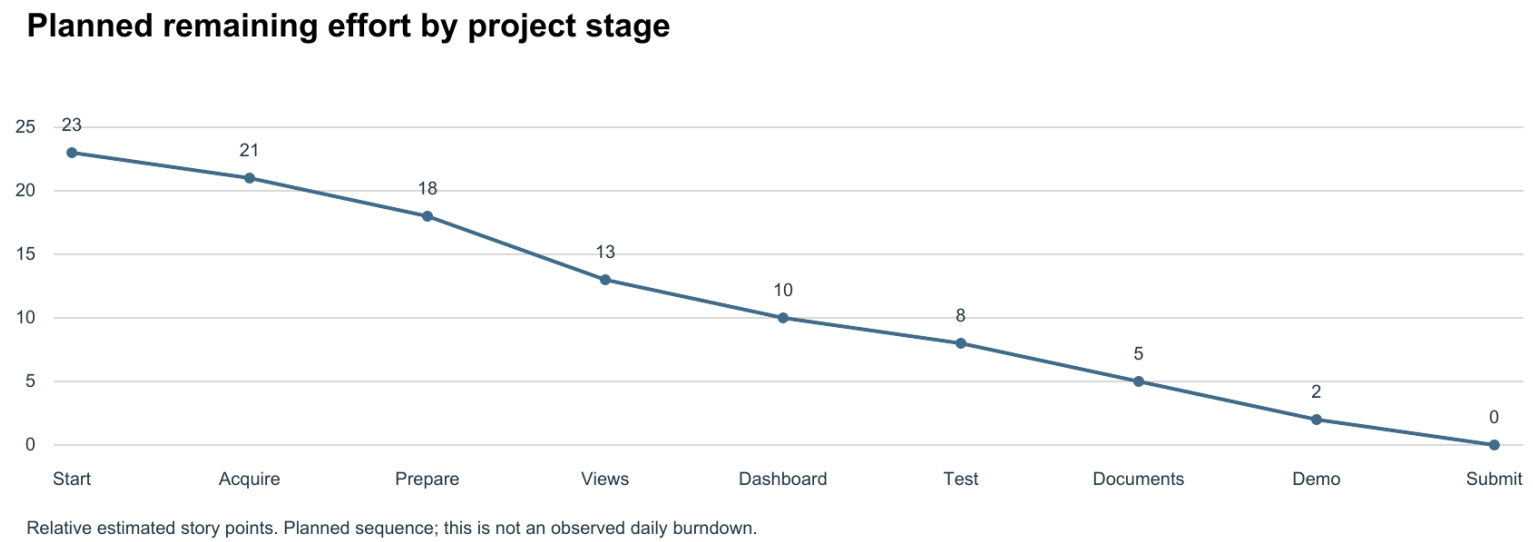
Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint 1	15	1 day	31 Oct 2026	31 Oct 2026	15	31 Oct 2026
Sprint 2	8	1 day	31 Oct 2026	01 Nov 2026	3	In progress

Planning assumptions

Story points are relative estimates for this individual capstone. Historical team velocity is not available. The tracker is a snapshot after completion of the narrated demo on 1 November: 18 of 23 estimated points completed, with documentation checks and submission in progress.

Planned burndown

The chart shows estimated remaining points by planned stage, rather than measured daily productivity. Final review is outside the development effort estimate.



Relative estimated story points. Planned sequence; this is not an observed daily burndown.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	5 Marks

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Snowflake Project – Data Engineering & SQL Solution

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	5 Marks

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission should demonstrate that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis..

Instructions

1. Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
2. The solution should address the project requirements through the required data preparation and transformation steps.
3. Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
4. Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

Solution Summary

Field	Details
Project / Repository Link	https://github.com/vishwaschoyal/ev-market-insights-tableau
Data Source(s) Link	Kaggle synthetic EV dataset; source link is provided in the final report.
Database	Not applicable to Tableau track
Warehouse	Not applicable to Tableau track

Field	Details
Tables / Views Created	Not applicable to Tableau track
Key Transformations / Calculations	Not applicable to Tableau track
KPIs created	Not applicable to Tableau track
Key Insights	Not applicable to Tableau track

Snowflake Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake	Not applicable to Tableau track
2	Data is cleaned, transformed, and prepared appropriately	Not applicable to Tableau track
3	Required tables, views, and data relationships are correctly implemented	Not applicable to Tableau track
4	Required SQL logic, calculations, and KPIs are correctly implemented	Not applicable to Tableau track
5	Data and solution outputs are validated against the project requirements	Not applicable to Tableau track
6	Solution is properly organized, documented, and functional	Not applicable to Tableau track

Additional Notes / Comments:

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	5 Marks

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

Activity 1: Interactive and visually appealing dashboards

Creating an effective Streamlit dashboard involves presenting business data stored in Snowflake through clear, interactive, and easy-to-understand visualizations. Students should design the dashboard so that users can quickly understand important KPIs, trends, comparisons, and business insights.

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	5 Marks

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1: Identifying and Handling Null Values

Identify the NULL or missing values present in the Snowflake dataset, analyze the affected columns, determine the appropriate approach for handling them based on the data and business requirements, apply the required query, and validate the cleaned data to ensure it is ready for further analysis and visualization.

Not applicable: Snowflake was not used. See the Tableau preparation and testing reports.

Activity 2: Duplicate Record Handling

Identify and analyze duplicate records in the Snowflake dataset, determine whether the duplicates are genuine or repeated entries, define the appropriate criteria for identifying duplicates, remove or retain records based on the project requirements, and validate the dataset after treatment to ensure that duplicate records do not affect the accuracy of further analysis and visualizations.

Not applicable: Snowflake was not used. See the Tableau preparation and testing reports.

Activity 3: Data Cleaning and Standardization

Identify and correct inconsistencies, invalid entries, formatting issues, and data quality problems in the Snowflake dataset, standardize values and formats across relevant columns, apply the required SQL-based cleaning and transformation steps, and validate the cleaned and standardized dataset to ensure consistency, accuracy, and readiness for further analysis and visualization.

Not applicable: Snowflake was not used. See the Tableau preparation and testing reports.

Activity 4: Data Transformation and Validation

Apply the required data transformations to prepare the cleaned data for analysis and modelling. Create necessary derived fields, apply business rules, filter or transform records where required, and validate the prepared data and other relevant quality checks before using it for the CURATED and MART layers.

Not applicable: Snowflake was not used. See the Tableau preparation and testing reports.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Business Question and Visualization Report

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

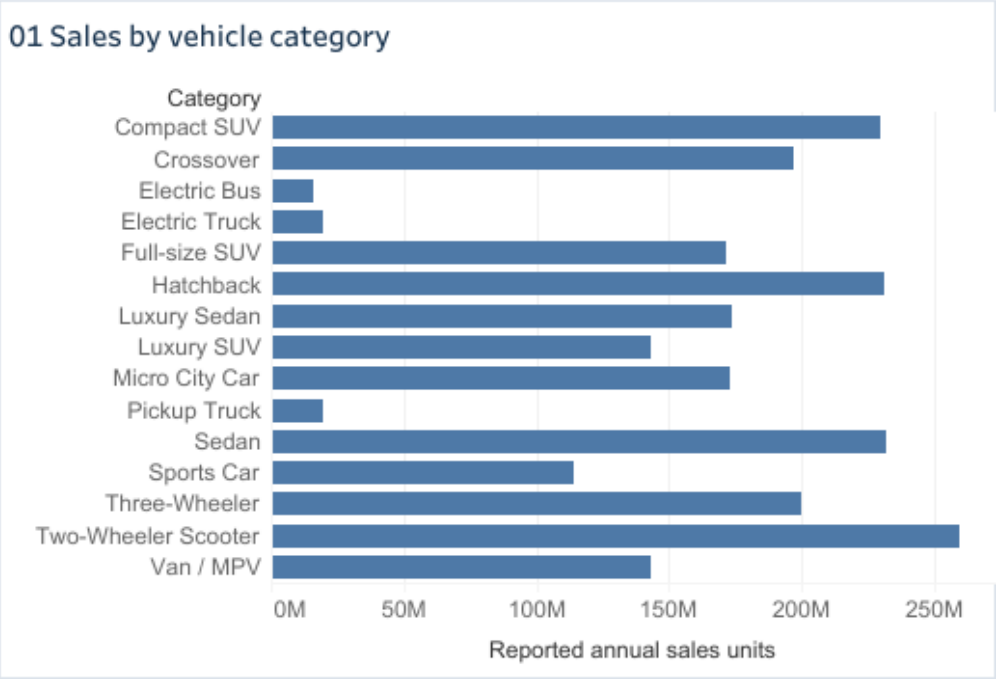
Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Visualization 1

Which vehicle categories contribute most reported sales?

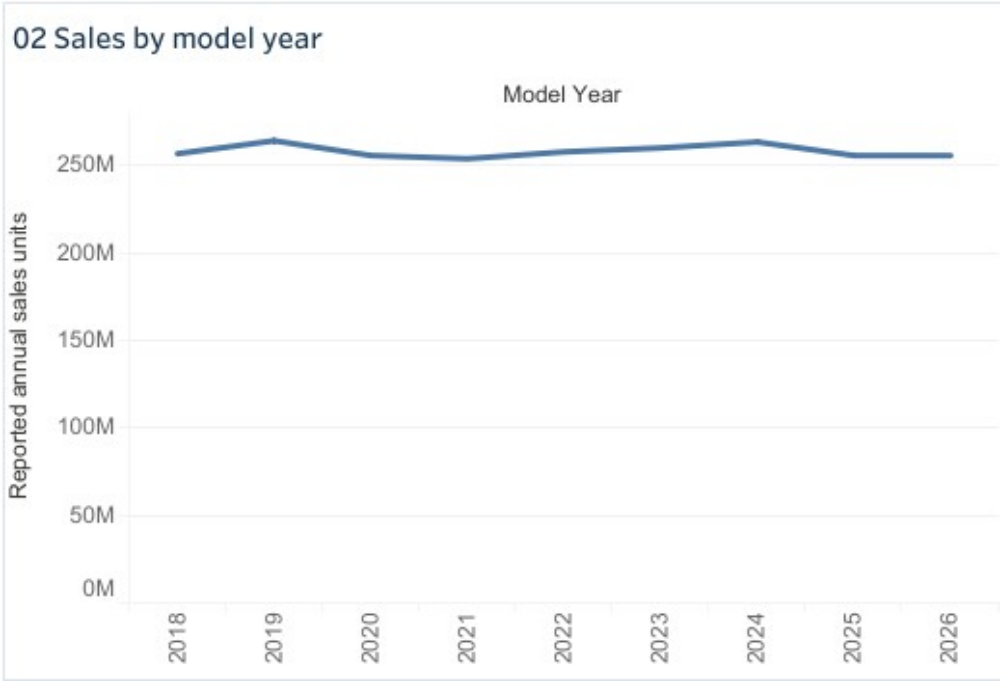
Visual: Bar chart. Two-Wheeler Scooter contributes the largest reported category sales, 259,647,705 units (11.2% of dataset sales).



Visualization 2

How do reported sales differ across model-year cohorts?

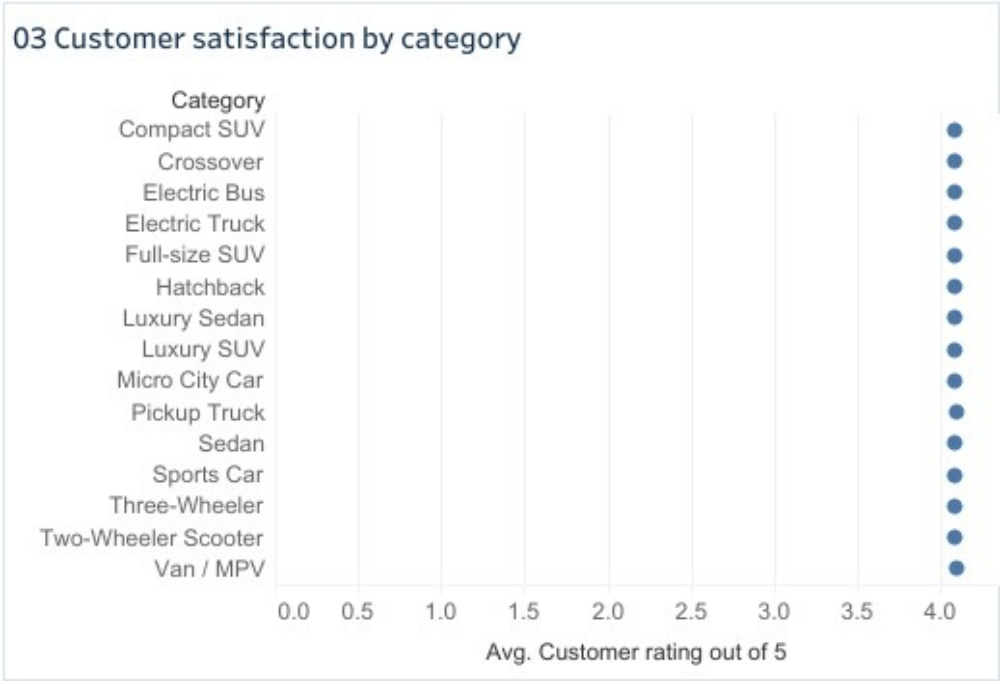
Visual: Line chart. Model-year cohorts span 2018 to 2026. Differences between cohorts are not evidence of transaction-year market growth.



Visualization 3

How do average customer ratings differ across vehicle categories?

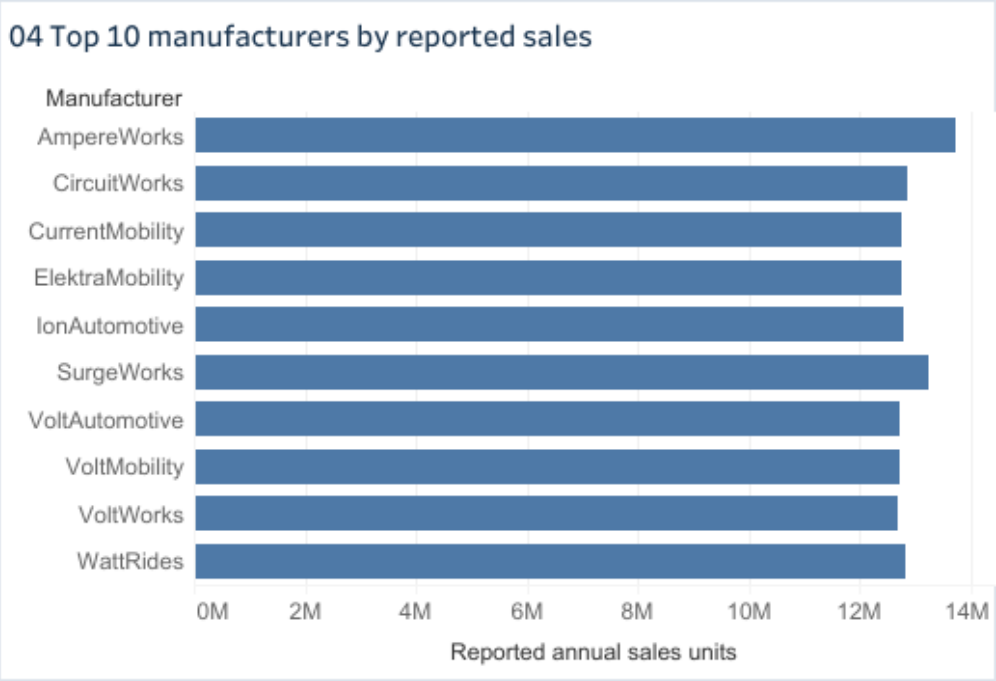
Visual: Circle chart. Pickup Truck has the highest mean category rating, 4.09 out of five, among records with observed ratings.



Visualization 4

Which ten manufacturers lead reported sales and what shares do they contribute?

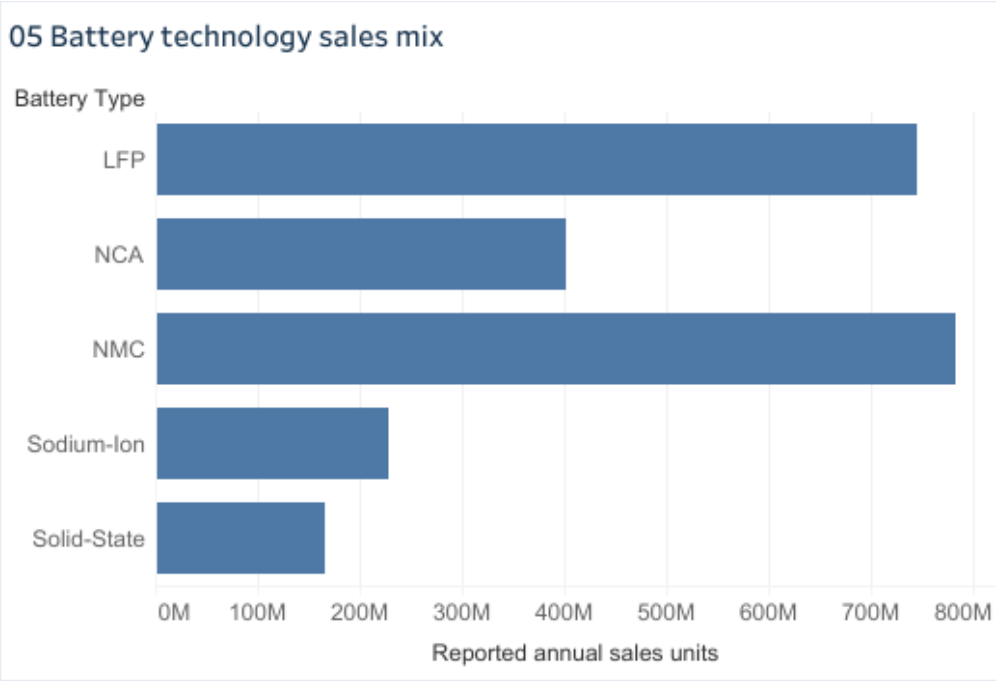
Visual: Bar chart. AmpereWorks leads reported manufacturer sales with 13,736,241 units (0.59% of dataset sales). Names are fictional.



Visualization 5

Which battery technologies account for the most reported sales?

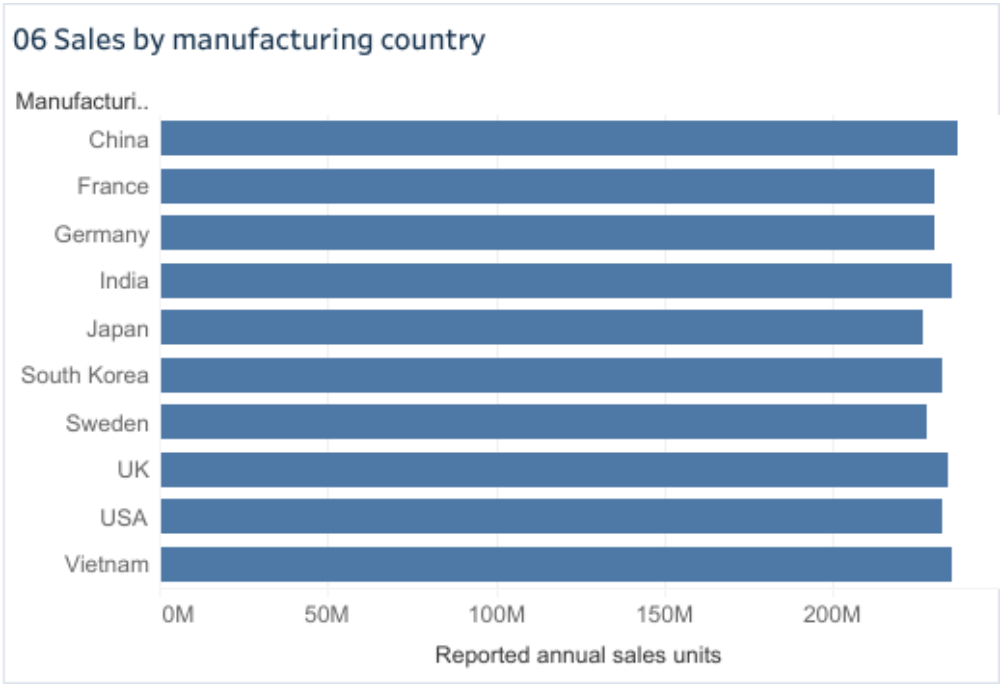
Visual: Bar chart. NMC leads battery-technology sales with 33.7% of reported dataset sales.



Visualization 6

Which manufacturing countries are associated with the most reported sales?

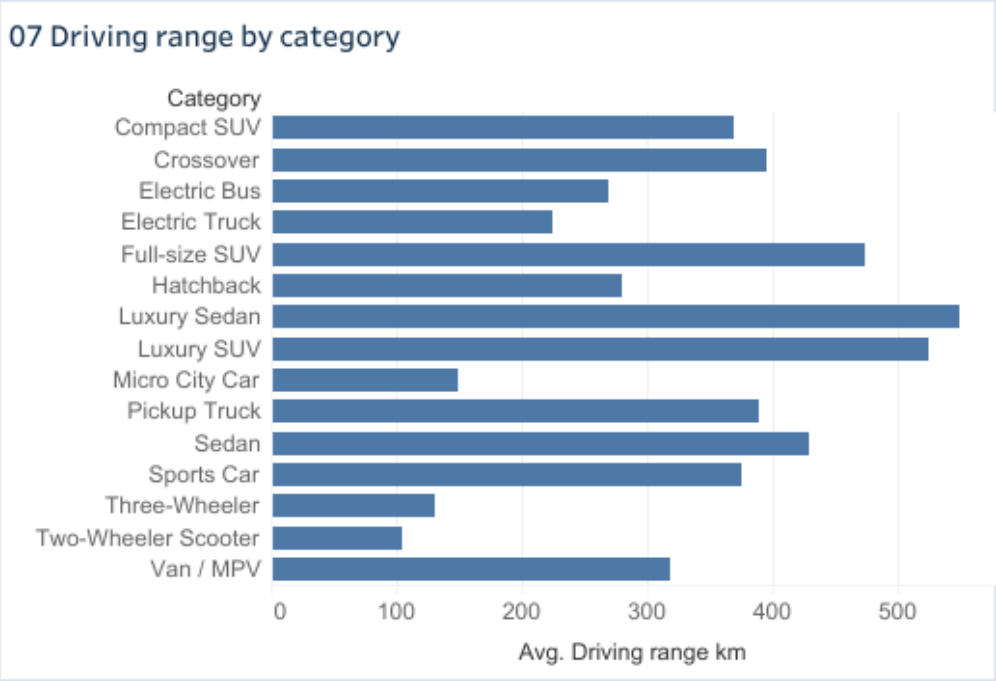
Visual: Bar chart. China leads sales associated with manufacturing country at 237,384,272 units. This measures neither factory output nor destination sales.



Visualization 7

How does average driving range differ across vehicle categories?

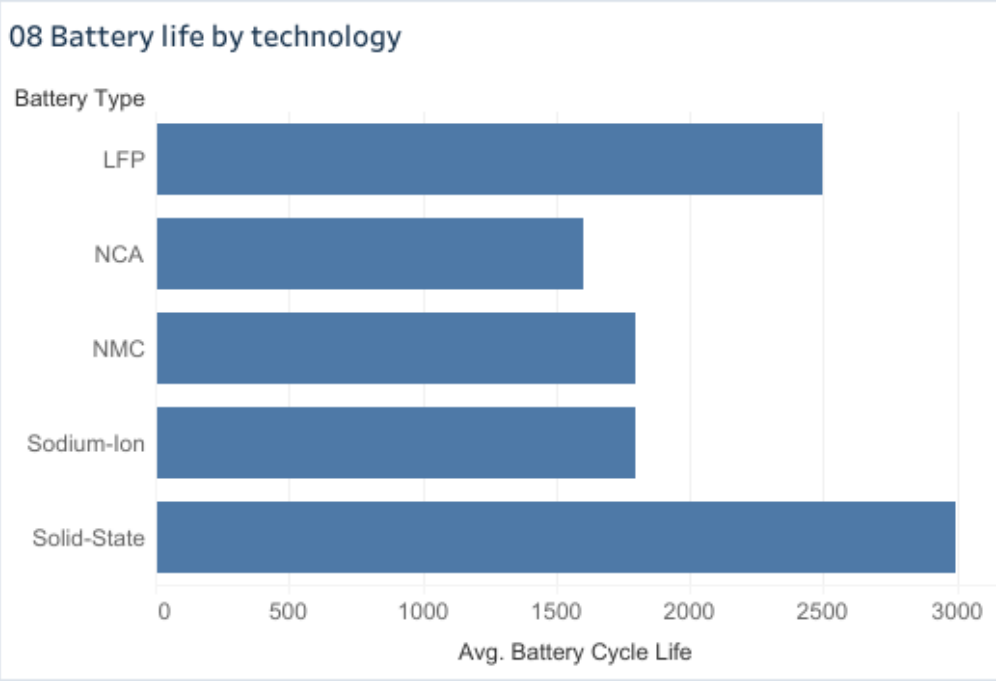
Visual: Bar chart. Luxury Sedan has the highest mean category range, 550.2 km. Different vehicle use cases limit direct comparisons.



Visualization 8

How do battery technologies compare in average cycle life and driving range?

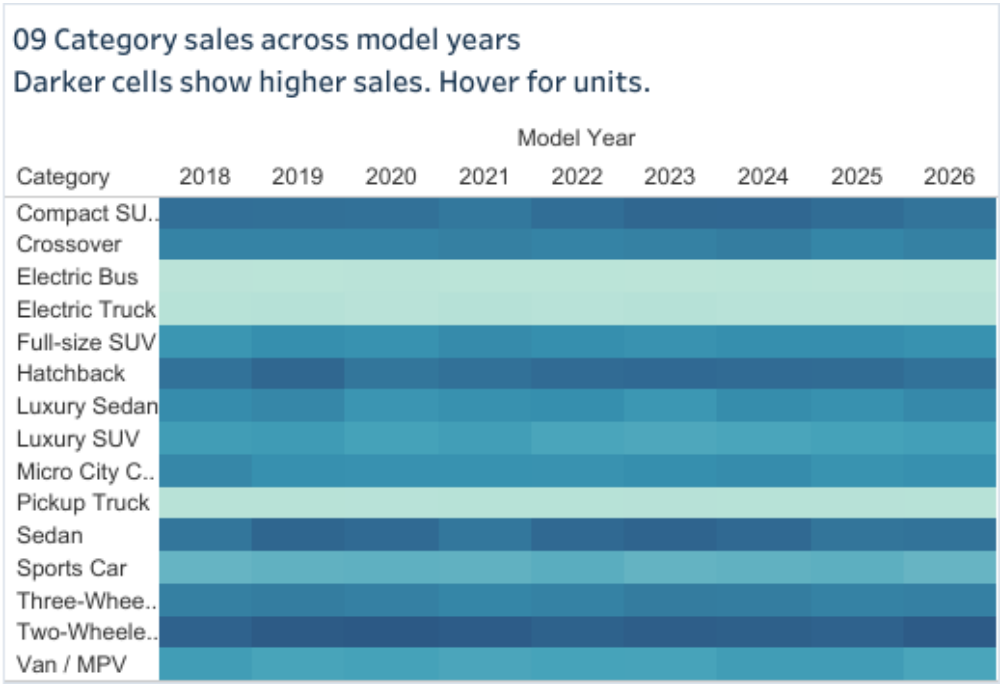
Visual: Bar chart. Average battery cycle life varies by chemistry. Compare cycle life in the bars and inspect average range and capacity in tooltips. These are synthetic technology characteristics.



Visualization 9

How does the category sales mix differ across model-year cohorts?

Visual: Square chart. The category-by-model-year heatmap highlights large sales cells and supports cohort composition comparisons. Missing sales remain unknown.



Interpretation limits

The publisher identifies this as synthetic data with fictional manufacturers. Sales by model year describe cohorts, and average ratings are unweighted source-record means.

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	5 Marks

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. Here are some key principles and best practices for dashboard design

Activity 1: Interactive and visually appealing dashboards

Creating interactive and visually appealing dashboards involves a combination of thoughtful design, effective use of visual elements, and the incorporation of interactive features. Here are some tips to help you design dashboards that are both visually appealing and engaging for users so take care of below points

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics

Implemented dashboard

A 1600 by 1200 canvas arranges nine views in three rows. The first chart is a category-selection filter. Tooltips expose ratings, sales and record counts. All chart panels use Entire View fitting.

The title identifies the synthetic data and record count. Model-year labels avoid implying transaction dates.

Two-Wheeler Scooter contributes the largest reported category sales, 259,647,705 units (11.2% of dataset sales).

Model-year cohorts span 2018 to 2026. Differences between cohorts are not evidence of transaction-year market growth.

Pickup Truck has the highest mean category rating, 4.09 out of five, among records with observed ratings.

AmpereWorks leads reported manufacturer sales with 13,736,241 units (0.59% of dataset sales). Names are fictional.

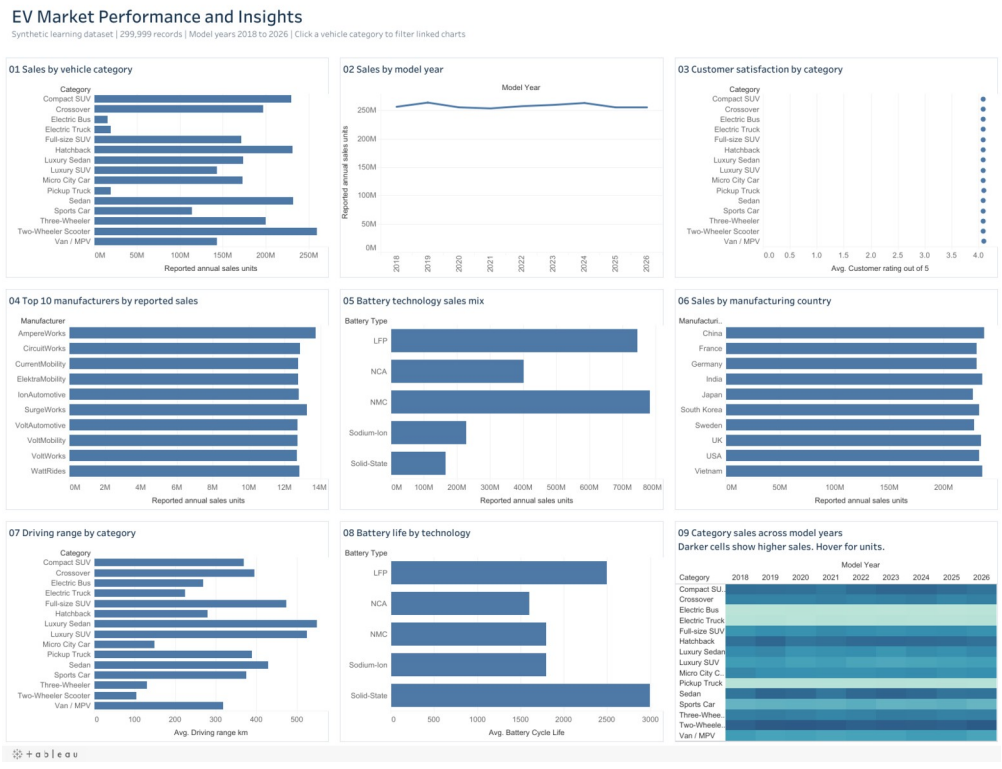
NMC leads battery-technology sales with 33.7% of reported dataset sales.

China leads sales associated with manufacturing country at 237,384,272 units. This measures neither factory output nor destination sales.

Luxury Sedan has the highest mean category range, 550.2 km. Different vehicle use cases limit direct comparisons.

Average battery cycle life varies by chemistry. Compare cycle life in the bars and inspect average range and capacity in tooltips. These are synthetic technology characteristics.

The category-by-model-year heatmap highlights large sales cells and supports cohort composition comparisons. Missing sales remain unknown.



Public dashboard

https://public.tableau.com/views/EVMarketInsights_17934697727980/EVMarketInsights

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	5 Marks

By using stories in Tableau, you can effectively communicate complex data in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It's a tool for **data storytelling**, allowing you to present insights in a cohesive, engaging way that takes viewers through a logical progression of findings or analyses.

In Tableau, **Story** is a feature that allows you to create a sequence of dashboards, visualizations, and text to present data insights in a cohesive and narrative-driven way. It's like a slideshow within Tableau that guides the audience through a series of data points, helping them understand key insights, trends, or outcomes of your analysis.

Scenario narrative

Story point 1 introduces Emily through category sales. The separate dashboard adds satisfaction, driving range and model-year cohort comparisons.

Story point 2 introduces David through the top manufacturers. Dashboard tooltips and linked views add sales contributions, battery mix and manufacturing countries.

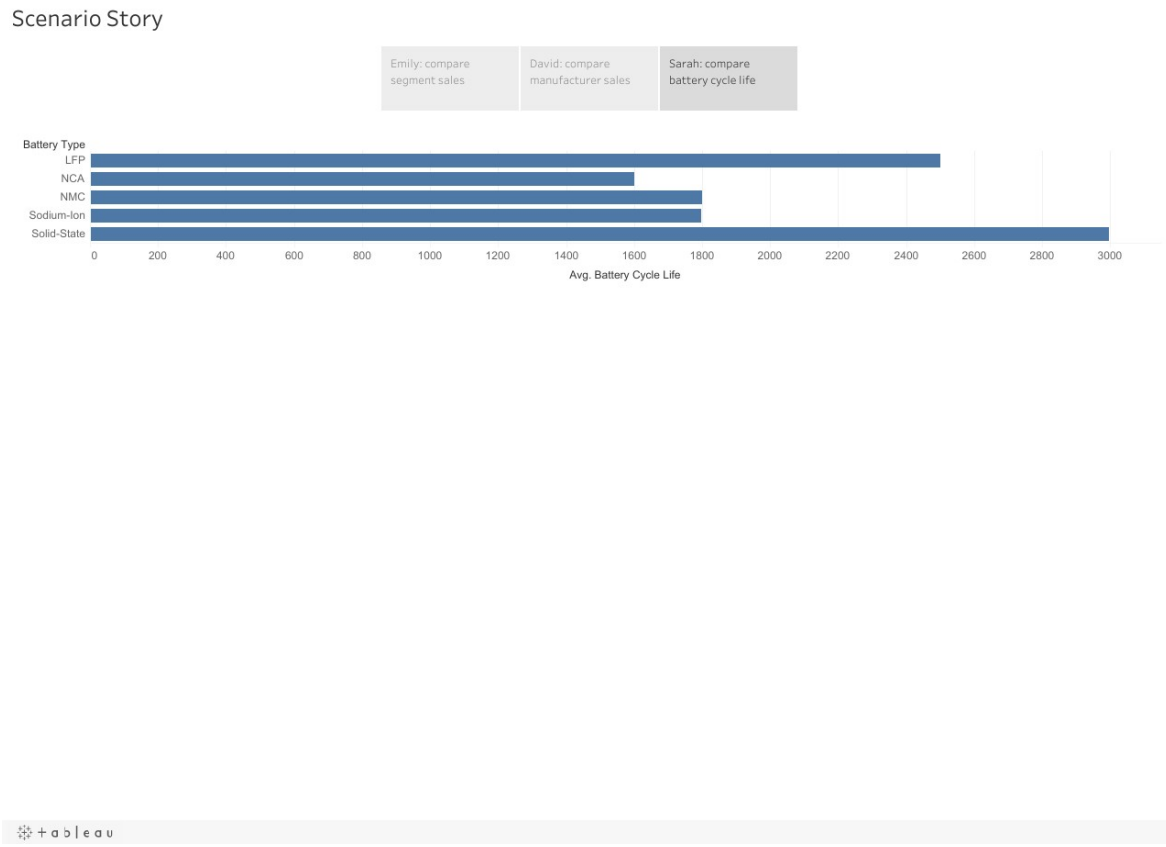
Story point 3 introduces Sarah through battery cycle life. Tooltips add average range and capacity; the separate dashboard adds category and geography comparisons.

Narrative conclusion

The dashboard connects commercial, technical and customer measures. The synthetic data supports an analytical demonstration, while real decisions would require measured market data and transaction dates.

Story publication

https://public.tableau.com/views/EVMarketInsights_17934697727980/ScenarioStory



Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Performance Test

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	

Track applicability: This project uses Tableau Public and a local Hyper extract. Snowflake SQL, ingestion and query-history evidence are not applicable. See the corresponding Tableau reports for implemented work.

Project Performance Testing:

The project team shall fill the following information in the Project performance testing template.

S.No.	Parameter	Screenshot / Values
1	Data Rendered	Not applicable to Tableau track
2	Data Preprocessing & Quality	Not applicable to Tableau track
3.	New Fields Created	Not applicable to Tableau track
4.	Filter Efficiency Test	Not applicable to Tableau track
5.	Dashboard & Visualization Testing	Not applicable to Tableau track
6	Query History	Not applicable to Tableau track

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Project Development Phase
Performance Test

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	

Project Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1	Data Rendered	299,999 cleaned rows; 37 source fields plus provenance and record key. Native Tableau load verified.
2	Data Preprocessing	One exact duplicate removed. Missing sales and ratings remain null; no invalid values in the selected range checks.
3	Utilization of Filters	Hatchback selection changed trend, battery and manufacturer charts; clearing restored full data.
4	Calculation fields Used	Record Count = 1. Sales Share uses SUM(sales) divided by the fixed overall dataset sum.
5	Dashboard design	9 visualizations. 1600 by 1200 dashboard with Entire View panel fitting.
6	Story Design	Three scenario-based narrative scenes; native story status is recorded in final documentation.

Reconciliation

Independent Python total: 2,323,303,050 reported sales units. Hatchback tooltip: 231,033,059 units and 25,713 records.
Filter behavior was observed in Tableau. No dashboard latency measurement is claimed; preparation runtime alone is not an interaction-performance test.

SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project author

Vishwas Choyal | EN25CS3011066 | Individual project

Communication

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	1 Mark

Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Solo check-in	Daily plan	Local checklist	Vishwas Choyal	Track tasks
2	Progress review	Per milestone	Project notes	Vishwas Choyal	Check evidence
3	Issue review	As needed	Validation log	Vishwas Choyal	Resolve defects
4	Mentor review	At submission	SkillWallet	Author; mentor	Request review
5	Demo rehearsal	Before recording	Tableau; script	Vishwas Choyal	Check clarity
6	Final handoff	Once	GitHub; report	Author; mentor	Share outputs

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	Missing source values	Keep nulls; document coverage.
2	Model year ambiguity	Describe cohorts, not transaction years.
3	Synthetic data scope	Label fictional data throughout.

Demonstration of Proposed Features

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Source preparation	Deduplicate CSVs	Implemented	Yes	299,999 rows
2	Nine charts	Scenario coverage	Implemented	Yes	Views 1 to 9
3	Category filter	Linked exploration	Implemented	Yes	Reset verified
4	Portable workbook	Embedded extract	Implemented	Yes	TWBX package
5	Public dashboard	Browser access	Implemented	Yes	Tableau Public
6	Narrated demo	English walkthrough	Implemented	Yes	See report link

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%

Project Demo Planning

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Problem and scope	Three scenarios; synthetic data	0.6	Vishwas Choyal
2	Data preparation	Rows, duplicate and missing values	0.6	Vishwas Choyal
3	Market strategy	Views 1, 2, 3 and 9	2.1	Vishwas Choyal
4	Competitive analysis	Views 4, 5 and 6	1.5	Vishwas Choyal
5	Technology research	Views 7 and 8; tooltips	1.1	Vishwas Choyal
6	Interaction and conclusion	Category filter; limits; links	0.7	Vishwas Choyal

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Introduce the project and personas.
2	Solution Overview	Explain data preparation and nine views.
3	Live Feature Demonstration	Walk through charts and filter action.
4	Q&A Session	Discuss limits and potential improvements.

Scalability & Future Plan

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Synthetic observations	No real-market inference	High
2	No transaction dates	No verified sales growth	High
3	Static extract	Manual refresh required	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User load	Public browser view	Monitor load; tune views
2	Data storage	Local Hyper extract	Incremental refresh pipeline
3	Performance	Aggregated chart marks	Benchmark larger extracts
4	Security	Synthetic public dataset	Private deployment for real data

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Validate with measured market data	After source approval	Real-market evidence
Phase 3	Add dated sales and refresh automation	After date fields exist	Verified temporal analysis
Phase 4	Compare battery metrics within category	After validation	Reduce mix-related bias

Team Involvement in Demonstration

Date	31 October 2026
Team ID	Individual
Project Name	Electric Vehicle Market Performance and Insights
Maximum Marks	1 Mark

Team Involvement in Demonstration:

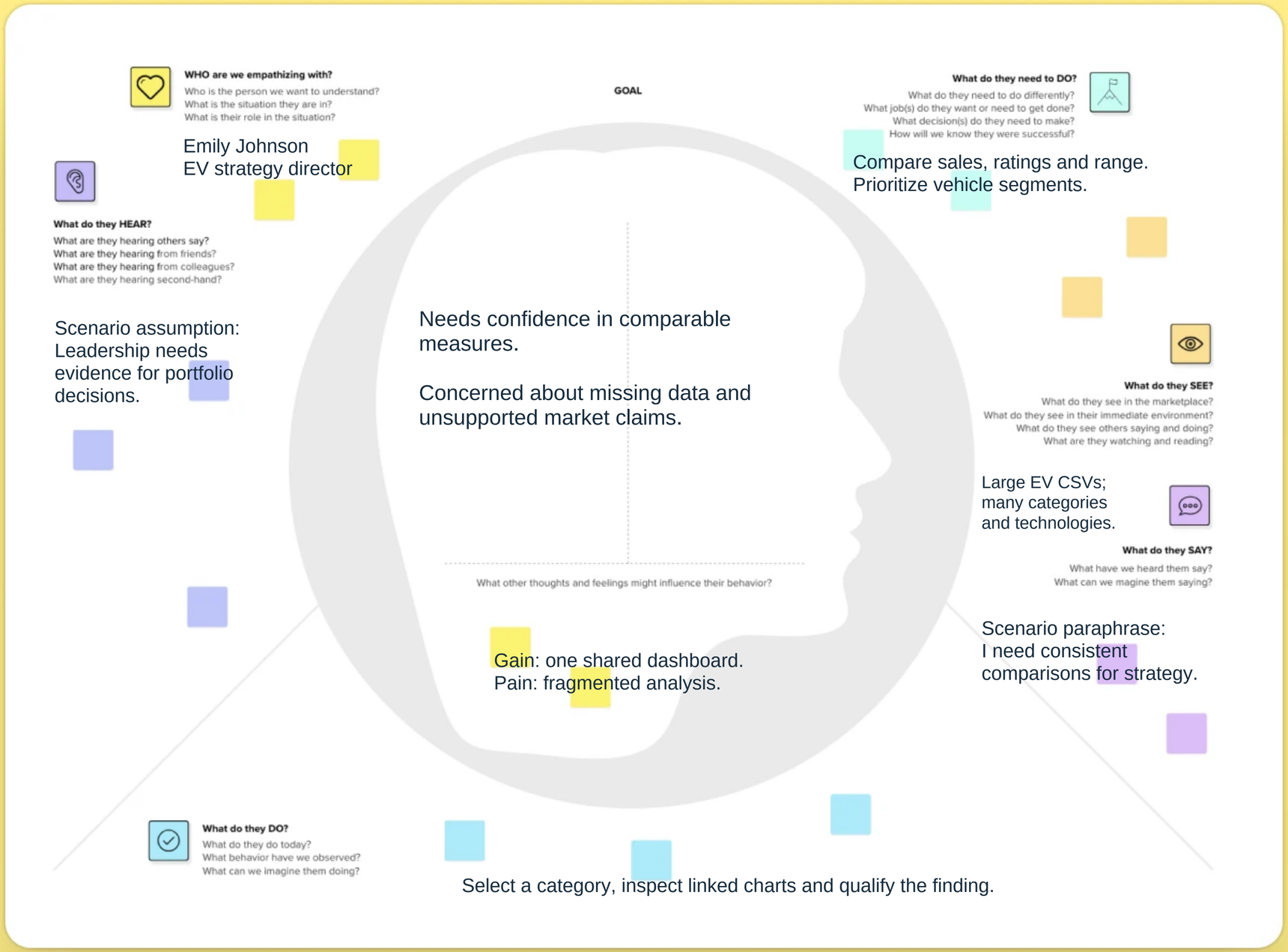
This document records the active participation and roles of each team member during the project demonstration. It ensures that every member contributes meaningfully to the presentation and that responsibilities are distributed fairly and clearly.

S.No	Team Member Name	Role in Demo	Section Presented	Contribution Summary	Participation (Active / Passive)
1	Vishwas Choyal	Individual author	All sections	Dashboard and documentation	Recorded
2	Not applicable	Solo project	Not applicable	Not applicable	Not applicable
3	Not applicable	Solo project	Not applicable	Not applicable	Not applicable
4	Not applicable	Solo project	Not applicable	Not applicable	Not applicable
5	Not applicable	Solo project	Not applicable	Not applicable	Not applicable
6	Not applicable	Solo project	Not applicable	Not applicable	Not applicable

Team Coordination Notes:

Aspect	Response
Team Leader / Coordinator	Vishwas Choyal
Overall Team Coordination Rating (1-5)	Not applicable to an individual project.
Any issues during demo	No recording failure observed.
How issues were resolved	Not applicable; the captured video was verified.

Emily Johnson empathy canvas



Scenario-derived assumptions, not interviews. Source template: Mural empathy map canvas.

EV project problem solution fit canvas

1. CUSTOMER SEGMENT(S)

EV strategy directors, BI managers and mobility researchers.

6. CUSTOMER STATE LIMITATIONS

Time, missing data and synthetic scope.

5. AVAILABLE SOLUTIONS

Raw CSVs and static spreadsheets; repeated manual comparisons.

2. PROBLEMS / PAINS + frequency

Compare demand, competitors, technology and ratings.
Recurring need at analytical reviews.

9. ROOT / CAUSE of every problem

Scattered raw data and inconsistent definitions.
No linked view across the questions.

7. BEHAVIOR + its intensity

Compare categories, inspect tooltips and reconcile totals before conclusions.

3. TRIGGERS TO ACT

Portfolio reviews and research questions.

10. YOUR SOLUTION

Nine Tableau views, category filtering, scenario story and documented validation.

© Daria Nepriakhina / IdeaHackers.nl

8. CHANNELS OF BEHAVIOR online + offline

Online: Tableau Public and GitHub.
Offline: TWBX and PDF reports.

4. EMOTIONS before & after

Uncertain before; clearer comparisons after.

Scenario-derived responses. Source: IdeaHackers canvas by Daria Nepriakhina; original attribution retained.